Type MS · Translators: assembler, compiler and interpreter

Answer 1 of 7 · Question 5 part a · 3 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct column:
<table>
<tr><th rowspan="2">Characteristic</th><th colspan="3">Translator</th></tr>
<tr><th>Assembler</th><th>Interpreter</th><th>Compiler</th></tr>
<tr><td>Does not have to be present in memory to execute the translated code.</td><td>✓</td><td></td><td>✓</td></tr>
<tr><td>Translates high level language source code.</td><td></td><td>✓</td><td>✓</td></tr>
<tr><td>Executes high level code until either the program has finished, or a syntax error is encountered.</td><td></td><td>✓</td><td></td></tr>
<tr><td>Translates low level language source code into machine code.</td><td>✓</td><td></td><td></td></tr>
</table>
</td>
<td></td>
<td>3</td>
</tr>
<tr>
<td>5(b)</td>
<td>Award two marks for a linked explanation such as:

• Fewer instructions are executed per second (1) so gameplay will not appear to execute as smoothly / graphics lag / fewer frames per second / can only handle less complex graphics (1)

• Consoles will execute more instructions per second / have faster processors / GPUs (1) so gameplay will not be realistic / mirrored / fair comparison (1)</td>
<td>One mark can be awarded processing speed implication, but the second mark is dependent on having a gameplay impact being identified.
e.g. “A lower clock speed means processing will take longer” is generic for one mark.

Low performance on its own is not enough.</td>
<td>2</td>
</tr>
</table>

16

Type MS · Translators: assembler, compiler and interpreter

Answer 1 of 7 · Question 5 part a · 3 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

<table>
<tr>
<td>5(c)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Issue</th><th>Utility software</th></tr>
<tr><td>Her word processor cannot read all of the files because it is corrupt.</td><td>File repair/recovery</td></tr>
<tr><td>She has a JPEG image and needs to send it to a publisher who can only read PNG image files.</td><td>Converting</td></tr>
<tr><td>There has been a flood and files on a server have been damaged, but Anna needs to access her data.</td><td>Backup/restore</td></tr>
<tr><td>Anna's hard drive is running very slowly when she tries to access a file.</td><td>Defragmentation/defrag</td></tr>
</table>
</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>5(d)</td>
<td>Award one mark for:
• Lossless (1)

Award one mark for any of the following up to a maximum of two marks:
• No data will be lost / the process is reversable (1)
• The manual will still be readable (1)
• Removal of data would change the meaning of text (1)
• Proofreader will be able to edit the text (1)</td>
<td>Allow converse if candidate gives reasons why lossy would not be used for justification

Allow RLE as a valid lossless compression method.</td>
<td>3</td>
</tr>
</table>

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Translators: assembler, compiler and interpreter

Answer 1 of 7 · Question 5 part a · 3 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

5(e)	Award **one** mark for each of: Method 1 • (C AND NOT D) (1) • (NOT C AND D) (1) • <expr1> OR <expr2> (1) Method 2 • (C OR D) (1) • NOT (C AND D) (1) • <expr1> AND <expr2> (1) Example statements: (C AND NOT D) OR (NOT C AND D) (C OR D) AND NOT (C AND D)	Order of precedence is NOT, AND, OR Note other equivalent solutions MAX 2 if all terms correct apart from missing brackets that are required. Do not credit circuit diagrams.	3
		Total for question 5	**15**

Type MS · Translators: assembler, compiler and interpreter

Answer 1 of 7 · Question 5 part a · 3 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

6(a)	Indicative content provided Other solutions do exist and should be awarded		6

Aspect of Solution	Marks			
	0	1	2	3
Functionality	No awardable content	There are significant errors in logic, leading to an overall solution that is non-functional	There are minor errors in logic, leading to an overall solution that is not completely functional	There are no errors in logic, leading to an overall solution that is fully functional
Accuracy of Notation	No awardable content	Notation follows a broadly unrecognisable convention that is applied inconsistently, although aspects of it are discernible	Notation follows a recognisable convention which is broadly discernible but is applied inconsistently	Notation follows a recognisable convention and is applied consistently throughout

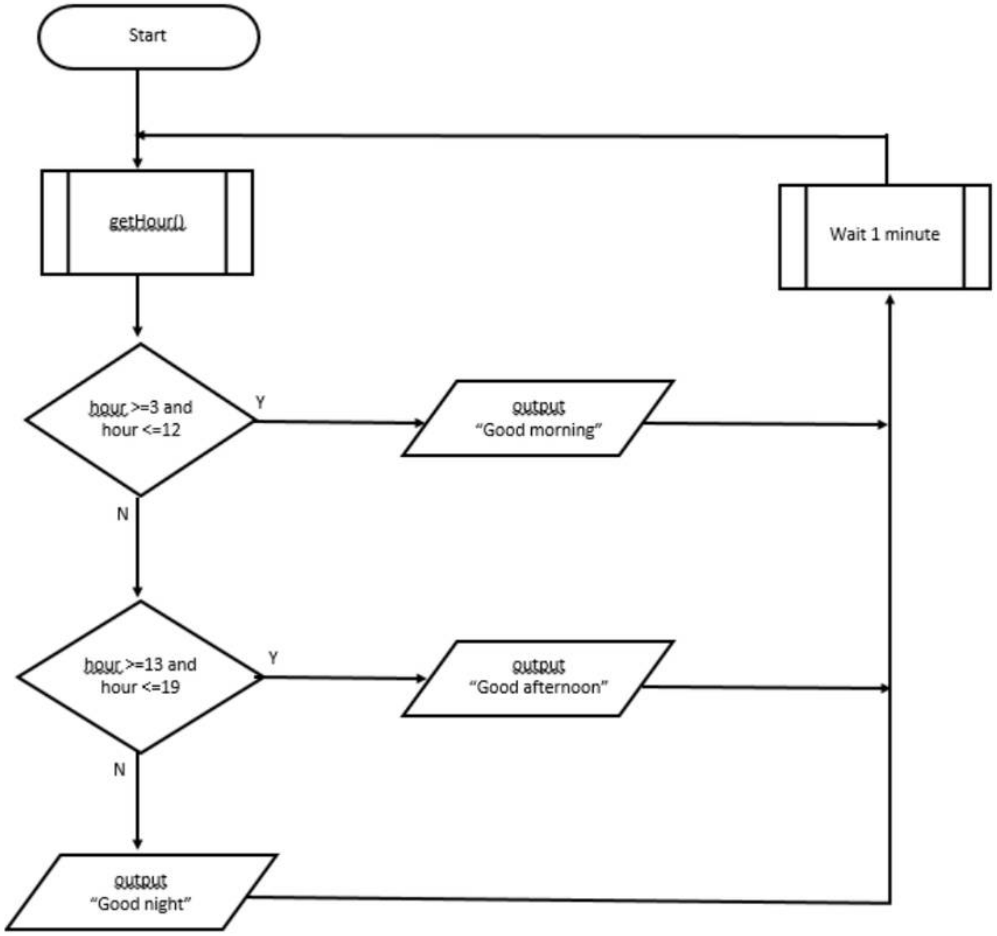
There are a maximum of 3 marks for functionality.
There are a maximum of 3 marks for accuracy of notation.
The marks for functionality and accuracy are awarded independently.

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Type MS · Translators: assembler, compiler and interpreter

Answer 2 of 7 · Question 6 part c · 6 marks

2020 NOV · Paper 01 · extract 2 of 8 · find part c in this mark scheme extract



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Type MS · Translators: assembler, compiler and interpreter

Answer 2 of 7 · Question 6 part c · 6 marks

2020 NOV · Paper 01 · extract 3 of 8 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	• papaya needs changing (1) • lychee needs ordering (1)		2

Question Number	Answer	Additional Guidance	Mark
6(b)(ii)	7 (1)		1

Question Number	Answer	Additional Guidance	Mark
6(b)(iii)	Replace 8 with LENGTH(flavours)/LENGTH(volume) (1) or A new line 7: WHILE i < LENGTH(flavours) DO (1) or WHILE i < LENGTH(volume) DO (1)	Ignore case If new line is given award mark if logic is clear	1

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Translators: assembler, compiler and interpreter

Answer 2 of 7 · Question 6 part c · 6 marks

2020 NOV · Paper 01 · extract 4 of 8 · find part c in this mark scheme extract

Question Number	Answer	Mark

20

Type MS · Translators: assembler, compiler and interpreter

Answer 2 of 7 · Question 6 part c · 6 marks

2020 NOV · Paper 01 · extract 5 of 8 · find part c in this mark scheme extract

6(c)	Indicative content: **Compiler** • One line of a compiled language maps to multiple executable instructions • Reads in a whole file and translates it at once • Produces an executable file • Executable file is portable between machines with the same architecture and operating systems • End-user cannot see the programming source code • Does not need an additional environment/software to run the code **Interpreter** • One line of a compiled language maps to multiple executable instructions • Reads, translates, and executes one line at a time • A special environment is needed to be installed on the user's machine to run the code; this is machine specific • Source code is portable to any machine with an interpreter which can run on it • End-user can see the programming source code **Similarities** • One line of a compiled language maps to multiple executable instructions/one to many relationship to machine code • Both are used to interpret high level programming languages • Both produce machine code from human-readable programming code **Differences** • Compiler reads in a whole file and translates it at once. Interpreter reads, translates and executes one line at a time • Compiler produces an executable file, interpreter does not • Executable files produced by a compiler will only execute on machines with the same architecture and operating systems. Source code produced by an interpreter is portable to any machine with an interpreter which can run on it • Compiled code does not require any additional software	6

Type MS · Translators: assembler, compiler and interpreter

Answer 2 of 7 · Question 6 part c · 6 marks

2020 NOV · Paper 01 · extract 6 of 8 · find part c in this mark scheme extract

	to execute the code. An interpreter needs a special environment to be installed on the user's machine to run the code; this is machine specific	

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Type MS · Translators: assembler, compiler and interpreter

Answer 2 of 7 · Question 6 part c · 6 marks

2020 NOV · Paper 01 · extract 7 of 8 · find part c in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Translators: assembler, compiler and interpreter

Answer 2 of 7 · Question 6 part c · 6 marks

2020 NOV · Paper 01 · extract 8 of 8 · find part c in this mark scheme extract

Type MS · Translators: assembler, compiler and interpreter

Answer 3 of 7 · Question 5 part c · 2 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

<table>
<tr><td>5(a)(i)</td><td>Award five marks for:

• Count column correct (1)
• Temp column correct (1)
• Row 2 Numbers[Count value from row above] swapped with Temp value (1)
• Row 3 Numbers[Count value from row above] swapped with Temp value (1)
• All correct (1)

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
<tr><td>1</td><td></td><td></td><td>10</td><td>3</td><td></td><td></td><td></td><td>10</td></tr>
<tr><td>2</td><td></td><td></td><td>6</td><td></td><td>9</td><td></td><td>6</td><td></td></tr>
</table>

<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
<tr><td>0</td><td>4</td><td>2</td><td>0</td><td>10</td><td>6</td><td>1</td><td>9</td><td>3</td></tr>
<tr><td>1</td><td>4</td><td>2</td><td>10</td><td>3</td><td>6</td><td>1</td><td>9</td><td>10</td></tr>
<tr><td>2</td><td>4</td><td>2</td><td>6</td><td>3</td><td>9</td><td>1</td><td>6</td><td>10</td></tr>
</table>
</td><td>Ignore any numbers that are greyed out</td><td>5</td></tr>
<tr><td>5(a)(ii)</td><td>Award one mark from:

• To reverse the contents of the array (1)
• To reverse the order of the numbers (1)</td><td></td><td>1</td></tr>
<tr><td>5(a)(iii)</td><td>Award two marks for a linked explanation such as:

• You need to swap the contents of array values (1) and without Temp one of the values would be lost (1)</td><td></td><td>2</td></tr>
</table>

Type MS · Translators: assembler, compiler and interpreter

Answer 3 of 7 · Question 5 part c · 2 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **two** marks for a linked explanation such as: • Isaac has included the -1 as a number in the addition (1) but the number has not been added to the count (1) • Isaac has misunderstood the WHILE loop (1) as it should not execute after the -1 has been input (1) • Isaac is expecting the wrong result (1) it should be 3 (1) • The count is 1 too many as the -1 is counted as a number (1) and the total is incorrect as 1 is subtracted from the total (1)		2
5(b)(ii)	Line 3 (1) Line 10 (1)		1
5(b)(iii)	SET count TO -1 (1) SET average TO (total + 1) / (count – 1)	Ignore brackets if the meaning is clear	1

Question Number	Answer	Additional Guidance	Mark
5(c)	Award **two** marks from: • Compiler produces object code to distribute that is difficult to reverse engineer / no need to distribute the source code (1) • Compiler optimises the code / object code (1) • Program runs faster (as it does not need translating) (1) • The target computer has no need to have the original compiler / does not need an interpreter (1) • Compiled code is smaller than the original code (1), which may not fit on the DVD (1)		2

Type MS · Translators: assembler, compiler and interpreter

Answer 3 of 7 · Question 5 part c · 2 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Translators: assembler, compiler and interpreter

Answer 4 of 7 · Question 1 part e.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part e.ii in this mark scheme extract

1(a)(i)	Award **one** mark from: • An action/task that is carried out (by the CPU) (1) • An operation/code that is/will be executed (by the CPU) (1) • Something that tells the CPU/processor/device what to do (1)	Allow examples	1
1 (a)(ii)	Award **one** mark from: • A pointer/reference to/identification of a location in memory (1) • A pointer/reference to/location where data will be accessed from/stored (to)(1)		1

Question Number	Answer	Additional Guidance	Mark
1(b)	**The only correct answer is D** *A is not correct because the address bus carries the address of the memory location* *B is not correct because the data bus carries but does not store data* *C is not correct because the control unit sends signals but does not handle the data*		1

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award **one** mark from: • More instructions can be carried out per second (1) • Processes run faster (1) • Programs executed faster (1) • Can run more complex programs (1)		1
1(c)(ii)	Award **one** mark from: • The CPU/computer could overheat (1) • More cooling required (1) • Needs more power (1) • CPU could become unstable/crash / its lifespan could be shortened (1)		1

Type MS · Translators: assembler, compiler and interpreter

Answer 4 of 7 · Question 1 part e.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part e.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(d)	**The only correct answer is D** *A is not correct because it describes a method of accessing data from a secondary storage device* *B is not correct because it describes the multi-agent computational model* *C is not correct because it describes the parallel computational model*		1

Type MS · Translators: assembler, compiler and interpreter

Answer 4 of 7 · Question 1 part e.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part e.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(e)(i)</td><td>Award one mark from:

• When fast execution of the program is essential/gives faster execution (1)
• When writing code to control a device / directly access the hardware (1)
• To make efficient use of a device's limited storage/memory/power (1)
• When writing code for a particular architecture/hardware/embedded system (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(ii)</td><td>Award one mark for:

• To translate/convert assembly language into machine/object code (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(iii)</td><td>Award one mark for any two correct rows (1)
Award two marks for any four correct rows (2)
Award three marks for all five correct rows (3)

<table><tr><th>Description</th><th>Compiler</th><th>Interpreter</th></tr><tr><td>Translates the program each time it is executed</td><td></td><td>☐</td></tr><tr><td>Produces permanent object code</td><td>☐</td><td></td></tr><tr><td>Translates line by line</td><td></td><td>☐</td></tr><tr><td>Translates the whole program before it is run</td><td>☐</td><td></td></tr><tr><td>Generates a list of errors once the complete program has been translated</td><td>☐</td><td></td></tr></table></td><td>Row can only be correct if there is only one tick.</td><td>3</td></tr>
<tr><td colspan="3">Total for question 1</td><td>11</td></tr>
</table>

Type MS · Translators: assembler, compiler and interpreter

Answer 4 of 7 · Question 1 part e.ii · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part e.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Translators: assembler, compiler and interpreter

Answer 5 of 7 · Question 1 part e.iii · 3 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part e.iii in this mark scheme extract

1(a)(i)	Award **one** mark from: • An action/task that is carried out (by the CPU) (1) • An operation/code that is/will be executed (by the CPU) (1) • Something that tells the CPU/processor/device what to do (1)	Allow examples	1
1 (a)(ii)	Award **one** mark from: • A pointer/reference to/identification of a location in memory (1) • A pointer/reference to/location where data will be accessed from/stored (to)(1)		1

Question Number	Answer	Additional Guidance	Mark
1(b)	**The only correct answer is D** *A is not correct because the address bus carries the address of the memory location* *B is not correct because the data bus carries but does not store data* *C is not correct because the control unit sends signals but does not handle the data*		1

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award **one** mark from: • More instructions can be carried out per second (1) • Processes run faster (1) • Programs executed faster (1) • Can run more complex programs (1)		1
1(c)(ii)	Award **one** mark from: • The CPU/computer could overheat (1) • More cooling required (1) • Needs more power (1) • CPU could become unstable/crash / its lifespan could be shortened (1)		1

Type MS · Translators: assembler, compiler and interpreter

Answer 5 of 7 · Question 1 part e.iii · 3 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part e.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(d)	**The only correct answer is D** *A is not correct because it describes a method of accessing data from a secondary storage device* *B is not correct because it describes the multi-agent computational model* *C is not correct because it describes the parallel computational model*		1

Type MS · Translators: assembler, compiler and interpreter

Answer 5 of 7 · Question 1 part e.iii · 3 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part e.iii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(e)(i)</td><td>Award one mark from:

• When fast execution of the program is essential/gives faster execution (1)
• When writing code to control a device / directly access the hardware (1)
• To make efficient use of a device's limited storage/memory/power (1)
• When writing code for a particular architecture/hardware/embedded system (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(ii)</td><td>Award one mark for:

• To translate/convert assembly language into machine/object code (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(iii)</td><td>Award one mark for any two correct rows (1)
Award two marks for any four correct rows (2)
Award three marks for all five correct rows (3)

<table><tr><th>Description</th><th>Compiler</th><th>Interpreter</th></tr><tr><td>Translates the program each time it is executed</td><td></td><td>☐</td></tr><tr><td>Produces permanent object code</td><td>☐</td><td></td></tr><tr><td>Translates line by line</td><td></td><td>☐</td></tr><tr><td>Translates the whole program before it is run</td><td>☐</td><td></td></tr><tr><td>Generates a list of errors once the complete program has been translated</td><td>☐</td><td></td></tr></table></td><td>Row can only be correct if there is only one tick.</td><td>3</td></tr>
<tr><td colspan="3">Total for question 1</td><td>11</td></tr>
</table>

Type MS · Translators: assembler, compiler and interpreter

Answer 5 of 7 · Question 1 part e.iii · 3 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part e.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Translators: assembler, compiler and interpreter

Answer 6 of 7 · Question 3 part f · 4 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part f in this mark scheme extract

3(a)	Award up to **two** marks for a linked explanation, such as: • If the data is incomplete/inaccurate (1) the answers from the model might not be right (1) • If the assumptions the model is based on are inaccurate (1) the answers from the model might be incomplete/inaccurate (1)	Award one mark for identifying a problem and one mark for a reason There must be a clear cause and consequence/effect	2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for any of: • Pixels per inch/ppi (1) • Pixels per centimetre/ppc (1) • Dots per inch/dpi (1)	Ignore units cm^2 $inches^2$	1

Type MS · Translators: assembler, compiler and interpreter

Answer 6 of 7 · Question 3 part f · 4 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part f in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	Award **two** marks for a linked description such as: • Each pixel/colour in the image (1) will require more bits to store (1)	Giving for formula file_size = width*height*colour_depth on its own is not enough	2

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked description such as: • System software is used to maintain/run the operation of the system/hardware (1) whereas application software allows the user to carry out a task (1)	For both marks both categories must be addressed Allow clear examples	2

Type MS · Translators: assembler, compiler and interpreter

Answer 6 of 7 · Question 3 part f · 4 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part f in this mark scheme extract

<table>
<tr><th>Question number</th><th>Answer</th><th>Additional guidance</th><th>Mark</th></tr>
<tr><td>3(d)</td><td>Award one mark for each of:

• All file blocks of each file are in numeric sequence/one after another in numeric order (1)

• All blank space is together at the front/at the end (1)

Examples:

<table>
<tr><td>W1</td><td>W2</td><td>W3</td><td>W4</td><td>Z1</td><td>Z2</td><td>Z3</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>Z1</td><td>Z2</td><td>Z3</td><td>W1</td><td>W2</td><td>W3</td><td>W4</td></tr>
<tr><td>Z1</td><td>Z2</td><td>Z3</td><td>W1</td><td>W2</td><td>W3</td><td>W4</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>W1</td><td>W2</td><td>W3</td><td>W4</td><td>Z1</td><td>Z2</td><td>Z3</td></tr>
</table>
</td><td>Note: Order of letter and number does not matter. W1 or 1W are equally acceptable</td><td>2</td></tr>
</table>

Type MS · Translators: assembler, compiler and interpreter

Answer 6 of 7 · Question 3 part f · 4 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part f in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)	Award up to **two** marks for a linked explanation such as: • The file size will not be decreased/will increase (1) because there are very few/no repeating patterns / because a run length of one would be added to each digit (1)		2

Question Number	Answer	Additional Guidance	Mark
3(f)	Award **one** mark for each correct cell:		4

Task	Translator
A guessing game that can be used on different computing platforms	Interpreter (1)
A screen driver for a new smartphone	Assembler (1)
A new version of a spreadsheet program for sale next year	Compiler (1)
Control software for an embedded system inside a new washing machine	Assembler (1)

Type MS · Translators: assembler, compiler and interpreter

Answer 6 of 7 · Question 3 part f · 4 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part f in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Translators: assembler, compiler and interpreter

Answer 7 of 7 · Question 5 part b.i · 1 mark

2025 JUN · Paper 01 · extract 1 of 5 · find part b.i in this mark scheme extract

5(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • Heart rate (1) to measure blood flow/pulse rate (1) • Barometric pressure (1) to determine altitude (1) • Microphone (1) for voice activation control (1) • Temperature/Thermometer (1) to measure ambient/body temperature (1) • GPS (1) to calculate longitude/latitude (1) • Light (1) to adjust screen brightness/charge watch battery (1) • Magnetometer (1) to act as a compass (1) • ECG (1) measure heart rhythm (1) • Blood pressure monitor (1) measure pressure in arteries / can report heart rate/pulse rate (1)	Allow any reasonable sensor that might be embedded within a smart watch. Purpose must be linked to the sensor Purpose must expand on name (e.g. heart rate sensor - monitor user's heart rate is not sufficient for second mark) Allow purpose mentioned in second column if nothing in first column	**2**
Question	Answer	Additional Guidance	Mark
5(a)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has a low power requirement (1) • It can use a low processor speed (1) • It has built in/integrated storage (1) • It relies on flash memory (1) • It can be mass produced at a low cost per device (1) • It is small (1) • It can work in a large range of operating temperature ranges (1) • It is robust (1) • It has its operating software set by the manufacturer (1) • It is designed to do one particular task (1)		**2**

Type MS · Translators: assembler, compiler and interpreter

Answer 7 of 7 · Question 5 part b.i · 1 mark

2025 JUN · Paper 01 · extract 2 of 5 · find part b.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark
5(b)(i)	**The only correct answer is B** *A is not correct because low-level languages are not converted into high-level languages* *C is not correct because assemblers translate assembly code to produce machine code but do not execute it* *D is not correct because compilers and interpreters translate high-level languages*		1
Question	Answer	Additional Guidance	Mark
5(b)(ii)	Award up to **two** marks for a linked explanation such as: • Makes best use of the hardware (1) because they can directly access (specific) hardware (features) (1) • It reduces manufacturing costs (1) because of a lower processor specification/expense (1) • Optimise their code for hardware (1) directly access/control registers (1)	Do not accept references to translation time	2
Question	Answer	Additional Guidance	Mark
5(b)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Abide by the law / avoid plagiarism / copyright infringements (1) • Adhere to a professional code of conduct (1) • Maintain/enhance their skills through further learning/keep up to date/attend (relevant) conferences (1) • Test code thoroughly/do not release code with known errors/vulnerabilities (1) • Respect privacy and security of data (1) • Preserve the environment (1) • Make software accessible/unbiased (1)		2

Type MS · Translators: assembler, compiler and interpreter

Answer 7 of 7 · Question 5 part b.i · 1 mark

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Question	Answer	Additional Guidance	Mark
5(b)(iv)	**The only correct answer is A** *B is not correct because 500 KiB is 512,000 bytes* *C is not correct because 0.5 MiB is 524,288 bytes* *D is not correct because 5MB is 5,000,000 bytes*		**1**
Question	**Answer**	**Additional Guidance**	**Mark**
5(c)	Indicative content: • Can use sensors to take/track readings such as heart rate • Can track location via GPS and record routes taken/distance travelled • Can calculate exercise/movement undertaken such as steps/calories burned • Can be linked via 5G/Internet Of Things/Personal Area Network (PAN) to other devices/apps that can record and analyse the data • Can trigger alerts when users are inactive • Can monitor blood oxygen/blood sugar/sleep patterns • Can applaud/give kudos when an achievement/event takes place • Can increase motivation of the wearer to be healthy/perform exercise due to feedback loop • Workouts/alerts can be synced to the watch • Linked apps can perform more in-depth analysis • No guarantee that users will buy into benefits and lead a healthier lifestyle • Monitoring vulnerable people allowing independent living	Allow content referring to other wearable technology which is not a smartwatch	**6**
		Total for question 5	**16**

Type MS · Translators: assembler, compiler and interpreter

Answer 7 of 7 · Question 5 part b.i · 1 mark

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Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Type MS · Translators: assembler, compiler and interpreter

Answer 7 of 7 · Question 5 part b.i · 1 mark

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Question	Answer	Additional Guidance	Mark